A Rigid-Flex Wearable Health Monitoring Sensor Patch for IoT-Connected Healthcare Applications

Taiyang Wu, *Student Member, IEEE*, Fan Wu, *Student Member, IEEE*, Chunkai Qiu, *Student Member, IEEE*, Jean-Michel Redouté, *Senior Member, IEEE*, and Mehmet Rasit Yuce, *Senior Member, IEEE*

***Abstract*—The Internet of Things (IoT) is a new communication paradigm that can connect elements from various fields through the Internet. One of the most attractive IoT applications is in the modern healthcare area, as the traditional healthcare system has an increasing demand for social resources, including doctors, nurses, hospital beds, and health monitoring devices. In this article, the design of a compact wearable sensor patch is presented for measurements of different physiological signals, such as the electrocardiogram (ECG), photoplethysmography (PPG), and body temperature. As ECG and PPG sensors are integrated with the same device, the proposed sensor patch can be used to estimate blood pressure (BP) continuously based on the pulse arrival time (PAT) without extra wires and devices. The sensor patch consists of a center board for signal acquisition and processing, a power board for energy supply and charging batteries, and three sensors for vital signs monitoring. All the components are designed in a rigid-flex structure, which can be easily attached to the human body for remote health monitoring applications. The sensors can be detached from the center board for customized measurements of a certain physiological signal (e.g., ECG) to reduce power consumption. Experiments are conducted to validate the performance of the proposed sensor patch by comparison with a commercial reference device. With the integration of a miniaturized Bluetooth low-energy (BLE) module, the proposed sensor system can transmit physiological measurements wirelessly to a gateway. Data encryption is applied on both the sensor patch and gateways to protect data for privacy and security purposes during transmission. Both a mobile gateway (based on smartphones) and a fixed gateway (based on portable computers) are designed as the bridge between the wearable sensor system and the Internet cloud, where health data can be stored and further analyzed. The experimental results demonstrate the feasibility of the overall platform for IoT-connected healthcare applications.**

***Index Terms*—Bluetooth low energy (BLE), electrocardiogram (ECG), Internet of Things (IoT), photoplethysmography (PPG), rigid-flex design, wearable healthcare.**

Manuscript received October 30, 2019; revised December 25, 2019 and February 5, 2020; accepted February 16, 2020. Date of publication February 28, 2020; date of current version August 12, 2020. This work was supported in part by the Australia Research Council (ARC) Linkage under Grant LP160101823. *(Corresponding author: Mehmet Rasit Yuce.)*

Taiyang Wu, Fan Wu, Chunkai Qiu, and Mehmet Rasit Yuce are with the Department of Electrical and Computer Systems Engineering, Monash University, Melbourne, VIC 3800, Australia (e-mail: mehmet.yuce@monash.edu).

Jean-Michel Redouté is with the Department of Electrical Engineering and Computer Science, Université de Liège, 4000 Liège, Belgium.

Digital Object Identifier 10.1109/JIOT.2020.2977164

I. Introduction

THE INTERNET of Things (IoT) has gained enormous attention from various research fields. Numerous elements from different areas can be connected with IoT technologies that can share information and resources without the limitation of time and distance [1], [2]. The modern healthcare is one of the most attractive IoT applications [3]. With the increasing human lifespan, chronic diseases are one of the predominant challenges to people's health worldwide [4]. The early diagnosis and treatment of chronic diseases can improve people's health conditions, which requires long-term continuous health monitoring of human vital signs, such as the heart rate (HR), respiration rate (RR), blood pressure (BP), body temperature, etc. [5]. A traditional healthcare system has limited social resources (doctors, nurses, hospital beds, and medical equipment) comparing with the increasing aging population [6]. The emergence of wearable devices with the IoT technology is a promising solution to the future health monitoring requirement.

In this article, a lightweight low-power compact wearable sensor patch is proposed for the monitoring of vital signs with IoT-connected healthcare applications, as shown in Fig. 1. The sensor patch consists of a center board, a power board, and three sensors, which are connected using flat flexible cables. Therefore, the sensor that is not required can be detached easily to reduce both the physical size and the power consumption of the system. The rigid-flex design of the sensor patch can be attached to the chest easily with bio-compatible tapes for measurements of physiological parameters, such as electrocardiogram (ECG), photoplethysmography (PPG), and body temperature. With the integration of ECG and PPG sensors, the continuous BP estimation can be achieved on the proposed sensor patch without extra setups.

All the sensor data will be transmitted through a Bluetooth low-energy (BLE) module to a gateway, which can be mobile (smartphones) or fixed (computers). AES-128 encryption is applied to protect the security and privacy of subjects during transmission. A timestamp will be inserted to each group of health data before the gateway forwards them to a cloud server for storage and further analysis. As mobility is not the focus of this article, a mobile gateway (smartphone) is utilized as a tradeoff solution to monitor users with movement requirements.

2327-4662 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

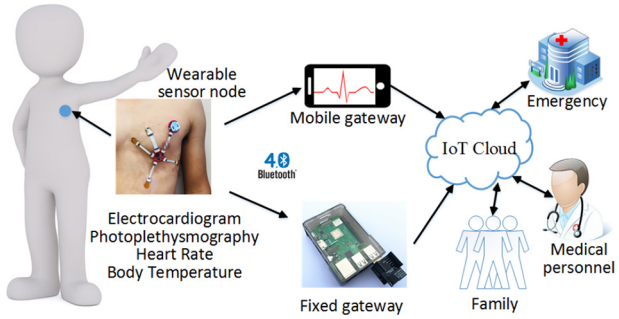


Fig. 1. Overview of the proposed wearable sensor patch system for IoT-connected healthcare applications.

The remainder of this article is organized as follows. Section II discusses and compares state-of-the-art wearable healthcare platforms with IoT applications. Section III describes the detailed hardware design of the sensor patch. The feasibility and performance of vital signs monitoring are presented in Section IV. In Section V, the implementation of the wireless network, cloud server, data encryption, and IoT applications for the proposed system are demonstrated. The summary and future improvement of the work are discussed in Section VI.

II. Related Works

An IoT-connected healthcare system usually consists of three main parts: 1) wearable sensors for vital signs monitoring; 2) a gateway to connect wearable devices to the Internet; and 3) a cloud server for data storage and further analysis. The gateway and the cloud server are the basic IoT infrastructure, while wearable sensors are indispensable components in an IoT-connected healthcare platform for remote health monitoring applications.

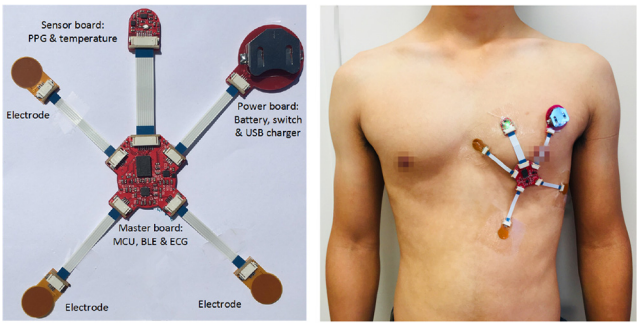
A. Wearable Sensors

The vital signs of the human body can reflect people's basic health status. There have been many wearable sensors proposed by researchers for measurements of health data. In [7], a comprehensive review of wearable sensors for remote healthcare applications is presented, such as cardiovascular monitoring, body activity and temperature, galvanic skin response (GSR), blood oxygen saturation (SpO_2), etc. For example, Asada *et al.* [8] proposed a wearable ring-shaped sensor for HR measurements based on the PPG, which analyzed technical and clinical issues during long-term continuous HR monitoring. Wearable motion sensors are employed for behavioral anomaly detection of the elderly in a smart assisted living home [9]. The proposed probabilistic framework based on wearable sensors can recognize anomalies of daily activities to improve the living status of elderly people. In [10], a small magnetometer-based sensor was designed to measure RR and the apnoea time during sleep. The sensor data are transmitted to a smartphone gateway and compared with a commercial airflow sensor. As for BP measurements, the traditional sphygmomanometer with a cuff is not suitable for wearable BP monitoring due to its cumbersome size and limited measurement intervals. In recent years, several promising methods for wearable BP monitoring are proposed based on the pulse transit time (PTT) and pulse arrival time (PAT). Most of the wearable BP estimation works measures ECG (on the body) and PPG (on the finger or earlobe) using separate devices [11]–[13], which is not user friendly for long-term monitoring scenarios. Thomas *et al.* proposed a wristwatch design for BP measurements in [14]. With the watch worn on the left wrist, a user just needs to touch the electrode with his right hand for ECG and PPG signal acquisition. The corresponding BP values can be calculated using an appropriate regression model for the PAT derived from the measured ECG and PPG signals. Although the biowatch simplifies the hardware design, it is not able to measure continuous long-term BP values as both hands are required for measurements. In the proposed work, both ECG and PPG sensors are integrated on a chest-based sensor patch, which can be used for long-term continuous BP estimation. Different from traditional wrist- and finger-based wearable devices, the proposed chest-based sensor patch can be hidden under clothes without disturbance in daily life activities.

B. Wireless Techniques and IoT Infrastructure

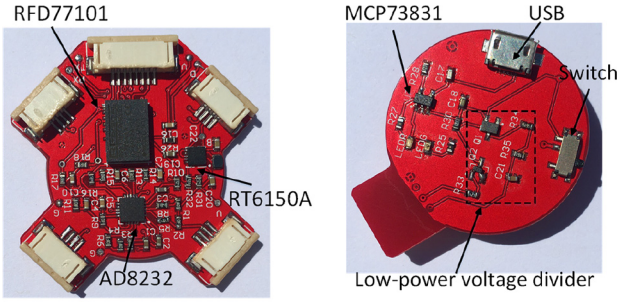
Wireless communication is a key technique to connect wearable sensors with IoT gateways. There have been several wireless protocols proposed for data transmission in short range (Zigbee, 6LowPAN, BLE, etc.) and in long range (LoRaWan, Sigfox, etc.) [15]. Researchers have proposed several wearable health monitoring systems for IoT-connected healthcare applications [16], [17]. For instance, the health-IoT platform is proposed with the integration of unobtrusive biosensors and an intelligent medicine box in [18]. A wearable bio-sensor is developed for ECG and body temperature measurements, while the medicine box is enabled with the RFID technology for patient identification and medication reminder. Gia *et al.* presented a low-cost wearable sensor node for the fog-assisted healthcare IoT system in [19]. The sensor node can collect and transmit ECG, RR, and body temperature to a gateway, which can provide notifications of emergencies. In [20], a personalizable wrist-worn ECG monitor is presented with an IoT infrastructure. The HR and HR variability (HRV) can be extracted from ECG signals, which are streamed to a smartphone through a BLE module. Mahmud *et al.* proposed a wireless health monitoring system using mobile phone accessories in [21]. The real-time ECG signal can be collected from the dry electrodes on the smartphone case, which will be stored and analyzed through a smartphone application for prognosis and diagnosis. However, it cannot record long-term ECG data as both hands are required for signal acquisition. In [22], an IoT-enabled wearable medical device was presented for measurements of multiple physiological parameters, such as HR, HRV, and RR. An Android phone is selected as the main gateway to store the local data and transfer them to the Internet cloud. However, using the smartphone as the sole gateway is not an optimal solution as the smartphone may be required for other daily usages. In this article, both a mobile and a fixed

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 2. (a) Rigid-flex design of the sensor patch for health monitoring. (b) Sensor patch is attached to a subject's chest using biocompatible tapes.



(a) (b)

Fig. 3. (a) Center board (42 mm × 40 mm) for signal acquisition, processing, and transmission. (b) Power board (⌀33 mm) for energy supply, voltage regulation, and charging batteries.

gateway are employed to receive and store physiological measurements in a local database and forward health data to an Internet cloud.

With the emergence of numerous IoT devices and platforms, IoT-related security issues are faced with new and unique challenges apart from the traditional context of the information technology. Khan and Salah [23] presented a thorough review of IoT security in terms of different security levels, protocols, standards, solutions, as well as future blockchain applications. As the IoT sensor nodes are resource-constrained (energy, latency, and computing power), some lightweight encryption methods are brought up or modified to meet new requirements [24], such as BCC [25], LEA [26], Twine [27], Skipjack [28], and AES [29]. In this article, the built-in AES-128 encryption engine of the BLE module is selected due to its low power, low latency, and low-computing requirements. Furthermore, as ECG is measured in the proposed sensor patch, the biometric ECG authentication can be a potential method for privacy protection in the future [30].

For an IoT-connected wearable healthcare infrastructure, the users' mobility needs to be considered for continuous health monitoring [31]. A design framework with mobility support in a wearable health monitoring system was presented in [32], which discusses and compares a number of conceptual and technological mechanisms. In [33], different requirements and solutions for mobility supported healthcare systems are presented, such as using a coordinator node [34] or a handover mechanism [35]. In the proposed wearable healthcare system, a mobile gateway (smartphone) is employed to be a tradeoff solution to the mobility requirement as it is beyond the scope of this article. When the sensor patch is disconnected from the mobile gateway and reconnected to the fixed one, an intermittent buffer is applied to store the health data during the connection [36]. In our future work, the received signal strength index (RSSI) can be utilized to connect the wearable sensor patch among different mobility supported gateways, as well as potential localization applications.

III. Hardware Architecture

For wearable biomedical sensors, they should be lightweight and flexible to be attached to the human body without causing extra discomfort. Wireless data transmission is a basic requirement as using wires to connect wearable sensors with external healthcare devices is inconvenient. When long-term health monitoring is necessary, the power consumption needs to be considered as well. In this article, a wireless wearable sensor patch is proposed for vital signs monitoring, as shown in Fig. 2(a). Different sensor boards are connected with the center board using flat flexible cables. The rigid-flex structure of the sensor patch makes it easy to be attached to the chest with biocompatible tapes, as shown in Fig. 2(b). The proposed wearable sensor patch incorporates a center board for signal processing and transmission, a power board for energy supply, and three sensors for measurements of ECG, PPG, and body temperature.

A. Center Board

Fig. 3(a) shows the design of the center board, which connects the other sensor boards for signal acquisition and processing. The core of the center board is a BLE module, the RFD77101 from Simblee, which is a high performance and professional grade Bluetooth 4.0 transceiver. It incorporates an ARM Cortex M0 processor and an embedded antenna within a miniaturized volume (10 mm × 7 mm × 2 mm). The built-in microcontroller (MCU) will perform initial signal processing on the collected physiological data, such as the real-time DC-removing, low-pass filtering, and band-pass filtering functions. The measured vital signs are transmitted through the BLE module to a gateway for remote health monitoring, data storage, and further analysis. The selected BLE module is suitable for wearable applications due to its small form factor and low-power specifications. Its peak transmission current is 8 mA and it has an ultralow-power sleep mode with the current less than 4 μA. The rest of the center board is a buck-boost converter (RT6150A) for voltage regulation and an analog front end (AFE) IC chip (AD8232) for ECG signal acquisition.

B. Power Board

A separate board is fabricated for the power supply of the wearable sensor patch, as shown in Fig. 3(b). The backside of the power board is used to install a 120-mAh rechargeable battery, LIR2450 from Multicomp. It is regulated at a constant voltage of 3.3 V by a low-cost efficient buck-boost converter, RT6150A from Richtek. To extend the lifetime of

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

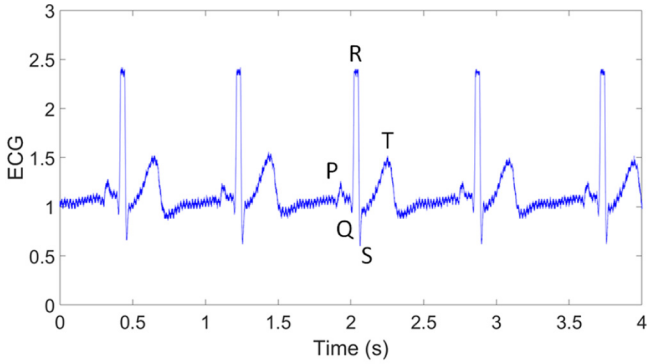


Fig. 4. Raw ECG signals collected by the ECG AFE circuit through the dry electrodes.

the rechargeable battery, a linear charge controller, MCP73831 from microchip, is exploited to manage the charging status of the battery. It first employs a fast charging mode with a constant current when the battery is in low capacity. After the battery reaches a preprogrammed voltage, the controller will charge the battery at a constant voltage with variable current.

To monitor the battery voltage status, an analog-to-digital converter (ADC) of the MCU on the center board is connected with the battery. Because the battery voltage may be higher than the MCU reference voltage (3.3 V), a low-power voltage divider circuit is implemented. Unlike the common two-resistor voltage divider which may drain unnecessary current from the battery, this voltage divider only works when the MCU gives it an "enable" signal (Bat-EN). Then, the P-channel MOSFET will be turned on and the MCU can read the battery voltage through the output (Bat-ADC) of a conventional voltage divider. When the battery measurement is not required, the MCU will pull down the Bat-EN and the P-MOSFET will be an open circuit (high impedance), thus there is almost no draining current from the resistors.

C. ECG Sensor

The foundation of the ECG sensor is the AFE circuit for ECG signal acquisition. In this article, an off-the-shelf ECG AFE chip, AD8232 from Analog Devices, is exerted for ECG data collection [37]. AD8232 is a fully integrated signal conditioning block that can extract, amplify, and filter small biopotential signals for ECG and other physiological measurements [38]. It can operate from 2 to 3.5 V with a low current of 170 μA. Its small size, low-power consumption, and low cost make it suitable for wearable healthcare applications. To minimize the distortion of ECG waveforms, a high-pass filter of 0.5 Hz and a low-pass filter of 40 Hz are used. The detected biopotential signals from electrodes are further amplified with the built-in operational amplifier to better display the ECG waveform.

For ECG biopotential acquisition, three dry copper electrodes based on flexible PCBs are printed and applied instead of traditional wet gel electrodes. The printed dry copper electrodes can be used for multiple times, which makes it more environmental friendly and cost effective than disposable wet electrodes [39]. The ECG AFE circuit can collect weak ECG signals through the dry copper electrodes with satisfactory quality. Fig. 4 demonstrates the raw measurement data of the ECG sensor from a 28-year old male subject using the DPO7104 oscilloscope from Tektronix. The basic features of ECG signals, such as the QRS complex, the P and T waves, and the ST segment can be clearly identified as shown in Fig. 4. The collected ECG signal will be further processed and transmitted wirelessly for real-time visualization, data storage, and future analysis on the IoT cloud.

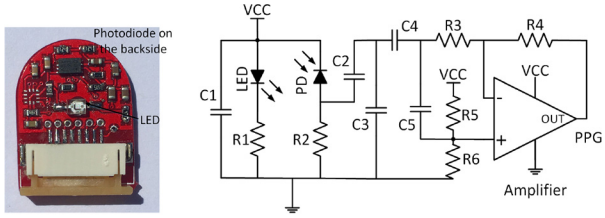


Fig. 5. (a) PCB design of the PPG sensor (20 mm × 16 mm). (b) Circuit diagram of the PPG sensor with an active low-pass filter.

D. PPG Sensor

Besides the ECG sensor, a PPG sensor is another common wearable device for human health monitoring applications. The PPG is a noninvasive optical technique for measurements of blood volume variations at the subcutaneous microvascular bed of tissues [40]. The basic theory of a PPG sensor is using a light source to illuminate skin tissues. Then, a photodiode (PD) is employed to detect intensity variations of the light transmitted through or reflected back by blood vessels and subcutaneous tissues. The transmitted or reflected light intensity can be changed by many substances of skin tissues, while the most dynamic variation is caused by the periodical blood flows. As the blood volume in vessels varies during the systolic and diastolic phase of a heartbeat, the light intensity variation detected by the PD is in the same frequency of the HR [41]. Based on the light from skin tissues, there are two types of PPG sensors: 1) the transmission mode and 2) the reflectance mode. The reflectance-mode PPG sensor is more commonly used in wearable applications as it can be placed at any place of human skin for sensing. In early PPG sensor designs, the red and infrared light is usually selected for HR and oxygen saturation measurements by transmission-mode PPG sensors. Recently, the green light is found to be a better option for reflectance-mode PPG sensors, as it has a better performance in the detection of variations of the superficial blood flows than the other light sources [42].

According to the requirements of the wearable sensor patch, a green light reflectance-mode PPG sensor is redesigned from our previous work [43], as shown in Fig. 5(a). It uses a small green LED to illuminate skin tissues for measurements. The reflected light is collected by a surface mounted PD, APDS9008 from Avago. The output signal from the PD is then processed by an active low-pass filter, as shown in Fig. 5(b). It will remove high-frequency noises and increase the amplitude of measured PPG signals. The PPG sensor is connected with the center board using a flexible flat cable, which is placed on

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

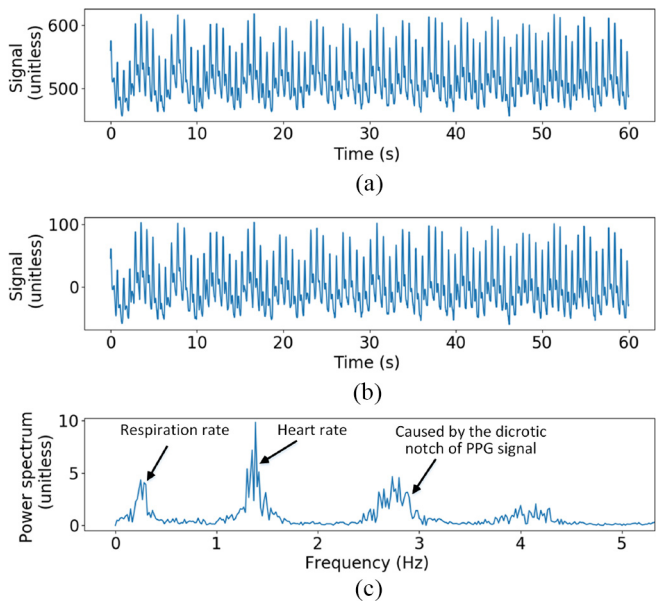


Fig. 6. Signals collected by the PPG sensor. (a) Original signals. (b) Signals after removing dc component. (c) Spectrum analysis of the PPG signals after removing dc component.

the chest for measurements. Fig. 6(a) demonstrates the original PPG signals of a subject measured on his chest for 1 min in a static position. After removing dc components [Fig. 6(b)], the fast Fourier transform (FFT) is applied for the spectrum analysis of the PPG data. The HR can be calculated from the time difference between PPG peaks. As the chest movement caused by breathing imposes a low-frequency signal on the dc component, it is possible to calculate the RR as well.

There is a high-accuracy temperature sensor, Si7051 from Silicon Labs, on the backside of the PPG board for body temperature measurements. All the sensors and the dry electrodes are connected with the center board using flexible flat cables. The rigid-flex design of the proposed sensor patch makes it convenient and comfortable to be attached to the chest for health monitoring. In addition, they can be detached easily from the sensor patch system. For example, the electrodes can be removed to make the sensor patch smaller when ECG monitoring is not necessary. Therefore, the sensor patch can be customized for measurements of the desired physiological parameters according to the user requirement.

E. Energy Consumption

In terms of wearable sensors, low-power consumption is an essential requirement for long-term continuous operation. In this article, the proposed sensor patch is implemented with low-power electronics. Table I shows the power consumption of the individual sensors, BLE, and MCU. To further reduce power consumption, the sensor patch can be configured in different operation modes for customized healthcare applications:

1) continuous measurements and transmission of all signals;
2) Continuous measurements of selected signals, cycled transmission of calculated vital signs;
3) cycled measurements and transmission of selected physiological parameters.

TABLE I
POWER CONSUMPTION OF THE PROPOSED WEARABLE SENSOR PATCH (VOLTAGE IS 3.3 V)

Item	Current	Power
PPG sensor	1.33 mA	4.39 mW
ECG sensor	176 μA	0.58 mW
Temperature sensor (@ 1Hz sample rate)	195 nA	0.64 μW
BLE (TX @ 0 dBm)	8 mA	26.4 mW
Microcontroller	4 mA	13.2 mW
Microcontroller (sleep)	7 μA	23 μW

The highest power of the proposed sensor patch is 44.57 mW (average current is 13.51 mA) when it collects and transmits all the sensor signals continuously for BP estimation on the gateway. In this mode, the sensor patch can operate up to 8.88 h with a 120-mAh battery. When only one sensor is required, the power consumption will be reduced. For example, if only PPG is activated for HR monitoring and sends HR data to the gateway every 1 min (BLE data transmission is less than 1 s), the power consumption can be reduced to 18.03 mW with an average current of 5.46 mA (1.33+4+8/60 mA). Therefore, the operation time of the sensor patch is 21.98 h when powered by the same battery. To further prolong the lifetime of the sensor patch, it can be configured into the cycled operation mode: 1-min normal operation and 4-min (can be configured according to different requirements) sleep. In this 5-min cycled operation mode, the sensor patch can operate for more than 100 h with the 120-mAh battery.

IV. HEALTH STATUS MONITORING

The different physiological parameters of the human body are basic reflections of people's health status. It is significant to keep a record of the physiological data periodically, especially for the elderly population. With the development of low-power miniaturized electronics and wireless communication techniques, various biomedical sensors have been proposed to monitor people's physical status for wearable and implantable healthcare applications. In this article, the proposed wearable sensor patch can measure ECG, PPG, and body temperature. Based on the ECG and PPG data, the HR and BP are estimated using appropriate signal processing models.

A. Electrocardiogram (ECG)

Among the multiple physiological parameters, the ECG signal is one of the most frequently monitored parameters as cardiovascular diseases (CVDs) are the principal causes of many catastrophic mortality [44]. The early diagnosis and treatment of CVDs will help a lot in improving patients' health status. Therefore, the real time and long-term monitoring of ECG is important in the prompt intervention of CVDs [45]. With the emergence of IoT technologies, remote ECG monitoring is achieved by transmitting data wirelessly to Internet cloud healthcare centers.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

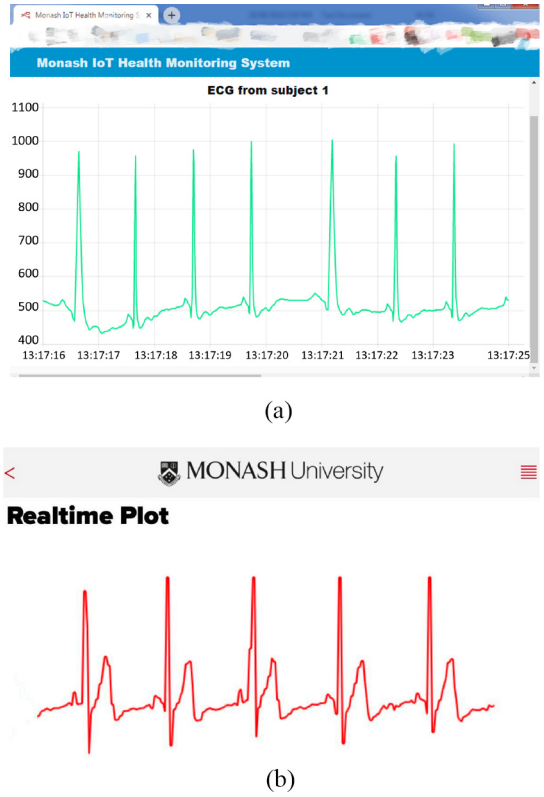


Fig. 7. (a) ECG waveform stored on the cloud and displayed on a website for remote monitoring. (b) ECG waveform displayed on a smartphone application.

In the proposed sensor patch, ECG signals are collected by the AFE circuit through the dry electrodes attached to the subject's chest. Then, the BLE module on the center board will transmit ECG signals to the gateway, which will forward the ECG data to an Internet cloud through the wireless network. The implementation of the wireless network and gateway will be presented in Section V. The ECG data will be stored in the cloud center for further analysis. The medical professionals with access to the cloud center can monitor subjects' ECG data remotely. As shown in Fig. 7(a), the ECG waveform can be visualized online by a website developed for the wearable sensor patch system. For the real-time ECG display, the data are sent to a gateway by the BLE module. The gateway can be a computer with the BLE connection, where ECG signals can be visualized on a computer graphical user interface (GUI) developed using Python. Another design of the gateway is based on a smartphone, which can display the ECG waveform on an application developed on the Evothings platform. Fig. 7(b) shows the screenshot of the smartphone application for the real-time ECG waveform.

B. Heart Rate

The heart is one of the most important organs, which functions as a pump in the circulatory system to provide oxygenated blood and other nutrients to the other body parts. HR is the cardiac muscle contraction and relaxation speed of the heartbeat. HR is one of the vital signs of the human body, which can be used as an independent risk factor of CVDs [46].

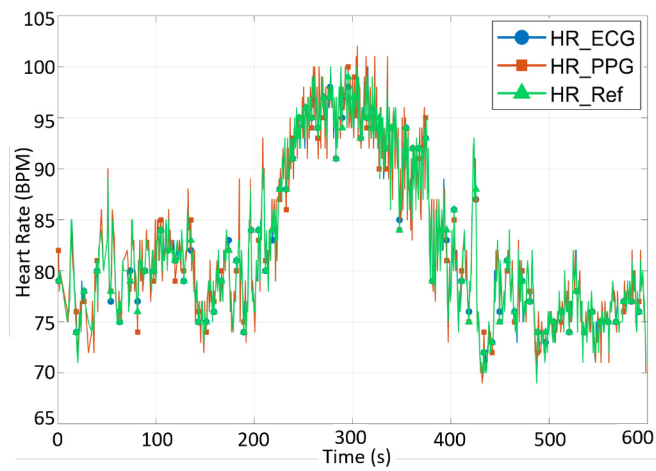


Fig. 8. HR measured by the ECG and PPG sensors versus the HR from a reference device.

TABLE II
HR CALCULATED FROM ECG AND PPG SIGNALS VERSUS REFERENCE VALUES ON TWO SUBJECTS (S1 AND S2): THE MEAN DIFFERENCE, STANDARD DEVIATION (SD), AND CONFIDENCE INTERVAL (CI)

		HR (ECG)			HR (PPG)		
		Mean	SD	CI (95%)	Mean	SD	CI (95%)
S1	Test1	-0.01	.044	[-0.88, 0.86]	0.04	1.38	[-2.67, 2.74]
	Test2	-0.01	0.46	[-0.91, 0.90]	0.03	1.48	[-2.87, 2.92]
S2	Test1	0.1	1.27	[-2.38, 2.59]	0.15	2.34	[-4.43, 4.73
	Test2	0.01	0.38	[-0.74, 0.75]	0.03	1.89	[-3.67, 3.74]

HR can be calculated from the *R–R* intervals of ECG signals or the time difference between peaks of PPG signals. Both the ECG and PPG sensors on the proposed wearable sensor patch are used for HR measurements, which are compared with a commercial health monitoring device, the Finapres NOVA with a 3-lead ECG module for HR monitoring. Fig. 8 demonstrates the comparison results among the HR values measured from the ECG sensor, PPG sensor, and the reference device. Both the sensor patch and the reference device are attached to the same subject for 10 min. The subject sits on a stationary bicycle and exercises for 3 min to increase his HR. As shown in Fig. 8, the HR values from the ECG and PPG sensors on the sensor patch have a high correlation (more than 98%) with the HR from the reference device. The same experiment is repeated twice on two male subjects (S1 and S2). Table II summarizes the comparison between the calculated HR from the sensor patch and the reference HR. The experimental results demonstrate that both the ECG and PPG on the proposed wearable sensor patch can provide reliable and accurate HR measurements, which will be monitored in the IoT-connected healthcare system.

C. Blood Pressure

BP is another vital sign of the human body, which is the force of the circulating blood pushing on the blood vessel

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

TABLE III
RMSE BETWEEN THE ESTIMATED BP AND REFERENCE BP OF SUBJECT-1 (S1) AND SUBJECT-2 (S2)

		Day 1	Day 2	Day 3	Day 4	Day 5	Overall
S1	SBP	9.16	6.06	6.14	7.41	4.43	6.64
	DBP	3.15	2.53	2.96	5.93	5.55	4.02
S2	SBP	6.82	5.78	7.68	5.03	6.92	6.45
	DBP	5.48	7.52	7.25	5.80	4.44	6.10

walls. BP is usually expressed in two numbers: 1) the systolic BP (SBP, the maximum value during a heartbeat) and 2) the diastolic BP (DBP, the minimum value during a heartbeat). The persistent high BP, also called hypertension, is a major risk factor of CVDs that may cause disability or even mortality. However, the diagnosis rate of hypertension is less than 46% because the early-stage symptoms are not clear [47]. Traditional BP measurements are taken using the sphygmomanometer with an arm cuff. It is not suitable for long-term continuous BP monitoring due to its cumbersome size and limited measurement frequency [48]. In recent years, several noninvasive BP estimation methods have been proposed by researchers for long-term continuous BP monitoring applications [49]. One of the promising techniques is based on the PAT, which is inversely correlated with BP [50]. The PAT is the time of a pulse wave propagating from the heart along the arterial tree to peripheral arteries. It can be derived from the time interval between the R peak of ECG signals and the maximum inclination point of PPG signals.

In our previous work [51], the ECG and PPG data from the proposed wearable sensor patch are applied to calculate the PAT. In order to convert the PAT to BP values, a regression model for the real-time BP estimation is proposed. Besides the PAT, HR is also correlated with BP, as HR will increase in an effort to raise cardiac output when BP rises. Therefore, the BP estimation model incorporates both the PAT and HR in a linear regression model

$$\mathrm{BP} = a \times \mathrm{PAT} + b \times \mathrm{HR} + c. \tag{1}$$

The coefficients (a, b, and c) are subject-specific model parameters, which can be calculated by the linear least square method against the reference BP values measured by a commercial beat-to-beat BP monitor, the Finarpes NOVA.

To find out the BP estimation model parameters, a 5-day experiment is conducted on two healthy male subjects in a lab room with a controlled temperature of 25 °C. Fig. 9 shows the experimental setup for data collection of the wearable sensor patch and the reference BP measurement device. The sensor patch is attached to the subject's chest for the acquisition of ECG and PPG signals, which will be transmitted by the BLE module to the computer for real-time signal processing. The noninvasive BP monitoring device, Finarpes NOVA is exerted for reference BP measurements. It can provide the real-time beat-to-beat BP values with a finger-cuff on the middle finger after an initial BP calibration procedure by an arm-cuff unit.

The data collection for the subjects lasts for five days. On each day, the subject is required to perform two 15-min training sessions for the calibration of the BP estimation model

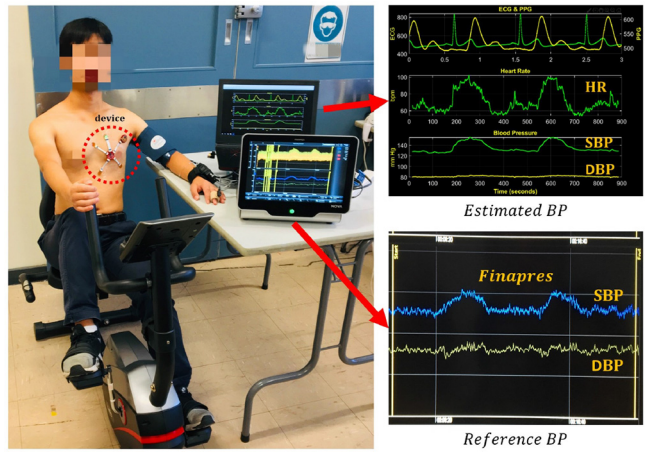


Fig. 9. Experimental setup for BP calibration and validation.

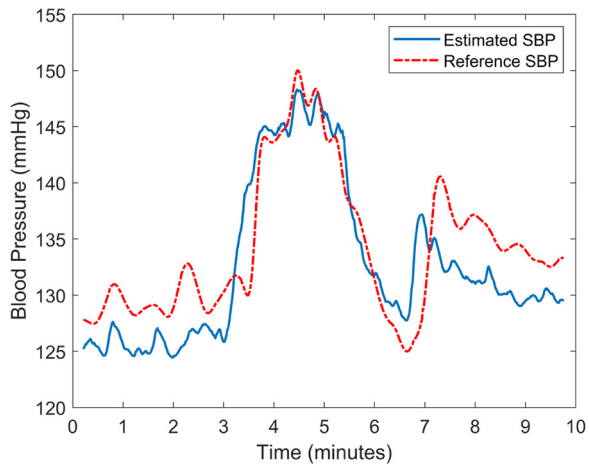


Fig. 10. Real-time BP estimation model based on the 5-day calibration data versus the reference BP measurements.

coefficients and one 10-min real-time validation session. There is a 10-min break between each session. During the training session, the subject sits on a stationary bicycle and performs two 3-min exercises to increase his BP. There is one 3-min resting period before and after the exercise to get his BP to a normal status. After the two training sessions, the coefficients are calculated and updated, which will be substituted to the BP estimation model. Then, the real-time validation session is carried out to verify the BP estimation model against the reference BP monitoring device. Table III summaries the root mean-square error (RMSE) of both subjects between the estimated BP and reference BP for each day and the overall experiment. The results show good agreement between the BP estimation model and the commercial device. Fig. 10 demonstrates the real-time SBP measurements based on the 5-day training model versus the reference SBP values, which has a strong correlation of 0.82. The experimental results demonstrate that the real-time BP monitoring based on the wearable sensor patch and estimation model is reliable given appropriate calibration procedures. It can be used for long-term continuous BP monitoring remotely in daily life.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

D. Body Temperature

Body temperature is usually regarded as the first vital sign, which is within the range from 36.2 °C to 37.5 °C. It requires a complex set of processes in the body to keep the temperature constant so that the other body functions can operate normally. Elevated body temperature usually implies a sign of infection or fever, which is the body defense mechanism's reaction to the stimuli of diseases. Therefore, it is important to measure body temperature periodically for the early detection of potential illness. In the wearable sensor patch, a miniaturized digital temperature sensor, Si7051 from Silicon Labs, is used for the periodical body temperature measurements. The sensor has high accuracy (0.1 °C) over its entire measurement range from 36 °C to 41 °C. Its industry-leading low-power consumption, average 195-nA current at the 1-Hz sampling rate, is suitable for wearable applications. The sensor patch will measure the body temperature every 30 s, which will be sent and stored in the cloud server. The real-time temperature data can be displayed on a smartphone application or on the website developed for the sensor patch system.

V. IoT-Connected Healthcare Platform

For IoT-connected healthcare applications, multiple wearable sensor patches are placed on different subjects for health status monitoring. The data from each sensor patch are transmitted through the BLE module to a gateway, which constitutes a wireless sensor network. The fixed gateway (with Wi-Fi connection) will forward the sensor data to an Internet cloud using a lightweight messaging protocol, the message queuing telemetry transport (MQTT). The medical professionals can visualize the health data from the subjects through a local user interface (UI) on the gateway or a cloud UI for remote healthcare situations. Besides the fixed gateway, a smartphone can be employed as the mobile gateway, which is connected to each individual subject's sensor patch. The personal health data can be displayed on a customized smartphone application and sent to the cloud server. All the sensor data can be viewed by healthcare staff with access to the cloud database. The cloud server will store multiple subjects' health data for backup and future analysis.

A. Implementation of Wireless Sensor Network

The wireless sensor network is implemented with BLE modules for local data transmission to the gateway, which will forward data to the Internet cloud using the MQTT protocol. BLE is an emerging wireless technology developed and marketed by the Bluetooth special interest group (Bluetooth SIG). BLE is designed to be efficient and low power while transmitting data at low latencies over a short range (0–100 m). Although there are other wireless technologies, such as ZigBee, 6LoWPAN, UWB, Z-Wave, Wi-Fi, etc., the usage of BLE is expanding rapidly due to its low-power consumption, low latency, and widespread adoption in commercial electronics (smartphones, laptops, automobiles, etc.) [52]. BLE has the potential to be a viable solution to various IoT applications [53], such as healthcare, environmental monitoring, smart home control, etc.

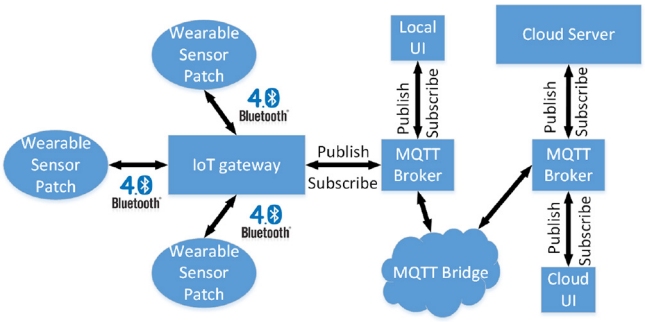


Fig. 11. Wireless network implementation for the IoT-connected healthcare system.

In this article, BLE is adopted to transmit health data from the wearable sensor patch to a fixed gateway in a local area (office room). It can be connected to a smartphone for data transmission and visualization when the subject is out of the fixed gateway coverage area.

To forward the wearable sensor patch data from different subjects to the Internet cloud, a messaging protocol that can work with limited bandwidth, power supply, and hardware resources is required. The MQTT protocol and the constrained application protocol (CoAP) have been proposed to address the specific requirements in the wireless sensor network and IoT applications [54]. However, CoAP is based on the one-to-one communication protocol for transferring state information between a client and a server, while the MQTT is a many-to-many communication protocol suitable for data exchange among different clients and brokers. Therefore, the MQTT is implemented on both the gateway and the cloud in this article. MQTT is a publish/subscribe messaging protocol that works on top of the TCP/IP protocol. It is designed as an extremely simple and lightweight messaging protocol for resource-constrained applications, such as low-bandwidth, high latency, and low-power sensors [55]. The normal MQTT consists of clients and brokers: clients can subscribe and publish messages to brokers on different topic while brokers handle the connections of clients [56].

Fig. 11 shows the implementation of the wireless network among the wearable sensor patches, the IoT gateway, and the cloud. Each wearable sensor patch has a unique MAC address, which can be used by the gateway to distinguish the data from different patches. The health data measured by sensor patches are transmitted to the IoT gateway through BLE modules. The sensor patch will receive an acknowledgment from the gateway when its data are received by the gateway. Once the data are received by the gateway, it will also be associated with a timestamp that can be used to synchronize data from different gateways. An MQTT broker is installed on the local IoT gateway, which is also configured as the MQTT bridge to connect to a cloud MQTT broker. The health data can be published to the local UI in the local area network for data visualization. Similarly, the remote health monitoring can be achieved on the cloud UI which is subscribed to the cloud MQTT.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

(a) (b)

Fig. 12. (a) Fixed gateway based on a tiny Raspberry Pi computer. (b) Mobile gateway based on a smartphone for data visualization.

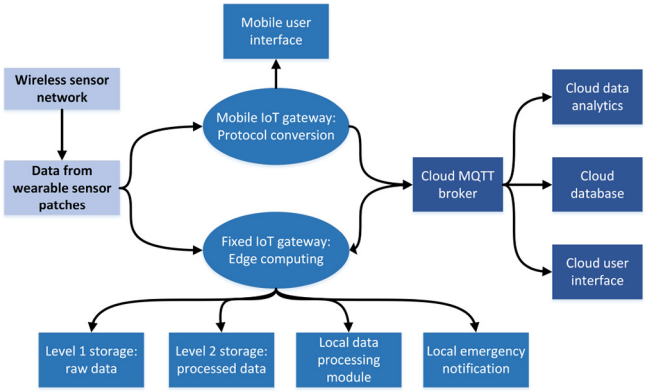


Fig. 13. Health data flow from the wearable sensor patch to the fixed gateway and IoT cloud.

B. IoT Gateway

The IoT gateway is responsible for connecting the local sensor network with the Internet cloud server, performing edge computing for signal processing, and hosting a local server for subjects' data visualization. To deal with different application scenarios (indoors and outdoors) of the wearable sensor patch system, both fixed and mobile gateways are designed and developed in this article.

The fixed gateway is implemented on a Raspberry Pi Model 3 B+, which is a low cost tiny single-board computer. It has both the Ethernet connection and Wi-Fi module onboard, which makes it a suitable platform for IoT applications [57]. The Raspberry Pi computers operate on a customized Linux operating system (OS) named Raspbian, an open-source OS with great community resources. The Raspbian system supports many of the existing programming languages, such as Python, Java, C, C++, Node.js, etc. [58]. Therefore, edge computing for preprocessing data and emergency notification can be realized easily with the support of multiple software platforms on the gateway. Edge computing refers to the technology that enables computing performed on the edge of a network before data are sent to the cloud [59], which has location awareness and low latency, and can reduce the pressure on the cloud computing [60]. Besides the abundant software support from the Raspbian system, the hardware of Raspberry Pi computers are suitable for IoT applications as well. For instance, the Pi series computers consume much lower power than a normal desktop or laptop. Its typical supply voltage and current are 5 V and 2.5 A, respectively, which means it can be powered by a portable power bank for a long time (a few hours) without the main power supply. Therefore, the gateway based on Pi computers can be relocated easily without shutting down the whole system.

In this article, the fixed gateway, the Raspberry Pi model 3 B+, is connected with the Simblee BLE module through one of its USB ports, as shown in Fig. 12(a). The separate BLE module is used to receive health data from different wearable sensor patches instead of the Pi's onboard BLE module to simplify the software programming functions required for edge computing. For data storage on the gateway, MySQL database is installed on the Raspbian system. Thus, health data from different subjects can be backed up and accessed when necessary. The Wi-Fi module on the Pi is exerted for the Internet connection to forward data to the cloud server. The MQTT is installed on the Pi as the messaging protocol between the gateway and the cloud. For local users, a lightweight Web UI based on Node.js is installed, which can be accessed by smartphone and Web browsers within the local area network.

When a subject wearing the sensor patch cannot be connected with the fixed gateway (out of range), a mobile gateway based on a smartphone can be employed as an alternative for receiving and forwarding data to the cloud. For sensor data visualization, a Web-based smartphone application is developed using the Evothings framework. It uses CSS, HTML5, and JavaScript code to build mobile applications, which does not rely on the platform-specific APIs of the Android or iOS system. The wearable sensor patch can be connected with smartphones through the BLE. The smartphone installed with the Evothings viewer can run the developed application to display data from the sensor patch, including the subject's ID, physiological data, and battery status, as shown in Fig. 12(b). The mobile gateway can be a supplement to the fixed gateway, which enables continuous monitoring of subjects' health status without space limitations.

C. IoT Cloud Server

The Internet cloud is an important component in various fields of IoT applications [61]. In this article, the cloud server is deployed in DigitalOcean, an American cloud infrastructure provider. The cloud server runs on the Ubuntu 16.04.5 system with 2-GB memory and 25-GB storage, which is sufficient for the present requirements. The cloud MQTT broker is installed on the cloud Ubuntu server to receive data from the IoT gateway. Once the data from different sensor patches are received by the IoT gateway, they will be forwarded to the MySQL database on the cloud for backup and further analysis.

D. Data Flow

Since there are two gateways (one mobile and one fixed) utilized in this article, it is of great essence to synchronize two gateways' data in the cloud and keep the local database up-to-date in order to ensure the data integrity. As mentioned

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

previously, each group of RF data is associated with a timestamp when received by the gateway. Therefore, the gateway and the cloud server can utilize the timestamp to synchronize the database.

Fig. 13 demonstrates the detailed data flow from sensor patches to the cloud server via the mobile and fixed IoT gateways. If the wearable patch is connected to the mobile gateway, the data will be visualized on a Web-based mobile application and transferred to the cloud via the MQTT protocol. The data received by the mobile gateway will also be transferred to the fixed IoT gateway so that the databases of both the fixed gateway and the cloud are synchronized. When the sensor patch is connected to the fixed gateway, the raw data are stored in the local MySQL database directly for backup (level 1 storage). Then edge computing, such as filtering duplicated data and providing real-time feedback, is performed to process the raw health data, which will be stored in the local MySQL database (level 2 storage) and forwarded to the cloud. The MQTT on the IoT gateway will publish the processed data to both the local server and cloud server. The local server can display sensor data on a Web browser within the local area network and send notifications when an emergency happens. The cloud server will store sensor data on the cloud database for backup and further data analysis. Medical professionals with access to the cloud server can view health data on the cloud UI, which enables the remote health monitoring application.

However, if the data are received by the mobile gateway rather than the fixed gateway, the data will be transferred to the cloud server via the MQTT broker. After the cloud server receives the data, it will compare with the data received from the fixed gateway. If they are duplicated from the same wearable sensor patch with the same timestamp, one set of the data will be filtered out. If the data are only received by the mobile gateway, these data will also be transferred to the fixed gateway via MQTT. In the local database, the backup data will be cleaned periodically to keep it up-to-date and make decision in time regarding emergency notification.

E. Security

Security and privacy issues are of critical importance when utilizing a gateway as a medium between sensors and cloud [62]. The issues need to be addressed and carefully considered in each layer of IoT platforms. For privacy, only the authorized person can have access to an individual person's health record. Since security is not the focus of this article, the two common security measures are implemented to meet the basic requirements of the following.

1) Security Between Wearable Patches and Gateways: Cryptography is one of the most common measures to protect sensitive data by converting data into ciphertext. Only the parties with the same key can decrypt the data. Therefore, the sensors and gateways with the same key can cipher and decipher the same information. Simblee is an advanced BLE module, which has a built-in AES encryption engine. Since the security is not the focus of this article, hence, the data communication between the wearable patches and the gateways are secured by the convention AES-128-0 encryption in this article. AES-128 does not require much memory and computational resources which is suitable for such resource-constrained wearable patches.

It is worth mentioning that, for resource-constrained devices like wearable sensor nodes, the data security can be implemented by other advanced lightweight cryptography methods that can further reduce processing power requirements, memory limitation, and energy consumption [63]. These advanced lightweight encryption methods for IoT applications, such as the BCC, LEA, and Twine, will be studied in our future work.

2) Security Measures in Gateways and Cloud Server: Since both the fixed gateway and the cloud server are running a distribution of Linux OS, a firewall called uncomplicated firewall (UFW) is installed in both systems. It can restrict the accessibility of different protocols and ports to reduce any access possibilities from an unauthorized person and achieve higher security levels. Therefore, only certain protocols can access the gateway and cloud server. For instance, in this article, UFW allows TCP connection from the MQTT broker via a specific port. Since the broker is deployed in our own server, only connections from desired IP ranges are permitted. Apart from firewall implementation, data encryption and decryption are also implemented in the gateway and cloud server to protect the data security.

F. Performance Evaluation

To validate the performance and feasibility of the overall IoT-connected healthcare system, three wearable sensor patches are tested on different subjects. The experiment is conducted in a lab room (14 m $\times$ 14 m) with a controlled temperature of 25 °C. The IoT gateway connected with Wi-Fi is placed at the center of the room to receive data. The sensor patches are attached to the chest of three subjects, who sit at different corners of the room. As ECG and BP measurements require active continuous data transmission and have been validated in Section IV, the ECG electrodes are detached from the sensor patch in this experiment. The PPG and temperature sensors are used to measure the HR and body temperature from three subjects. All the data from the sensor patches are transmitted to the IoT gateway, which forwards them to the cloud server. The data transmitted from the sensor patch to the gateways are acknowledged every time, otherwise, it will be sent repeatedly for three times. If the receiving acknowledgment is still not received, a miss sending counter is created, which will be added in the next sending cycle. For the data transmitted between the gateways and the cloud server using the MQTT protocol, the QoS is set to level 1 (at least once) so that we can guarantee each data are successfully transmitted to the cloud at least once. As shown in Fig. 14, the website based on the cloud UI can display the real-time HR and temperature of each subject for remote health monitoring. When ECG and BP measurements are required for serious subjects, the electrodes can be added to the sensor patch for ECG and BP monitoring.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF RECENT STATE-OF-THE-ART WEARABLE SENSOR SYSTEMS TOWARD IOT-CONNECTED HEALTHCARE APPLICATIONS

	[18] (2014)	[19] (2017)	[64] (2017)	[21] (2017)	[20] (2018)	[22] (2018)	This work
Microcontroller	——	ATMega328P	ATMega328P	ARM Cortex M0	ARM Cortex M3	ARM Cortex M3	ARM Cortex M0
ADC resolution	——	24-bit	24-bit	——	16-bit	24-bit	12-bit
Wireless technology	Wi-Fi, RFID & Zigbee	Customized 2.4GHz nRF24L01 transceiver	Bluetooth 2.0	BLE	BLE	BLE	BLE & Wi-Fi
Sensors	Self-designed SoC	ADS1292 & BME280	ADS1299	Self-designed ECG circuit	Self-designed ECG circuit	ADS1292	AD8232, Si7051 & customized PPG
Physiological parameters	Heart rate & body temperature	ECG, respiration rate & body temperature	ECG, EMG	Heart rate	Heart rate & heart rate variability	Heart rate & respiration rate	Heart rate, blood pressure & body temperature
Sensor location	Chest	——	Portable	Hand	Wrist	Chest	Chest
Dimension (mm)	45 * 25	——	57 * 38	Smart phone case size	50 * 47	40 * 20	42 * 40 (Centre board)
Power consumption (mW)	——	Idle: 1.2 Active without encryption: 19.5 Active with encryption: 21.03	Idle: 218.57 - 226.70 Active: 451.30 - 481.27	——	Active: 29.7 Sleep: 0.037	Normal: 10.53 Stop: 0.505	Continuous: 44.57 Cycled: 18.03 Sleep: 0.023
Power supply	——	1000 mAh battery	1700 mAh battery	5500 mAh battery	100 mAh battery	300 mAh battery	120 mAh battery
Operation time (hours)	——	up to 155	up to 13.6	——	48 days (30 s operation every one hour)	up to 85	up to 100 (1 min operation every 5 min)
Gateway	——	Orange Pi	Raspberry Pi	Smartphone	SPHERE Gateway	Smartphone	Raspberry Pi & smartphone
Data encryption	Yes	Yes	Yes	——	Yes	——	Yes
Visualization	PC application	——	PC Matlab application	Smartphone application	——	Smartphone & web application	Smartphone & web application

In recent years, different designs of wearable devices with temperature, HR, RR, and BP. Table IV presents a detailed IoT-connected applications have been proposed by researchers summary and comparison of recent wearable healthcare worldwide for health monitoring of vital signs, such as body systems. Most of the devices are designed to be attached

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

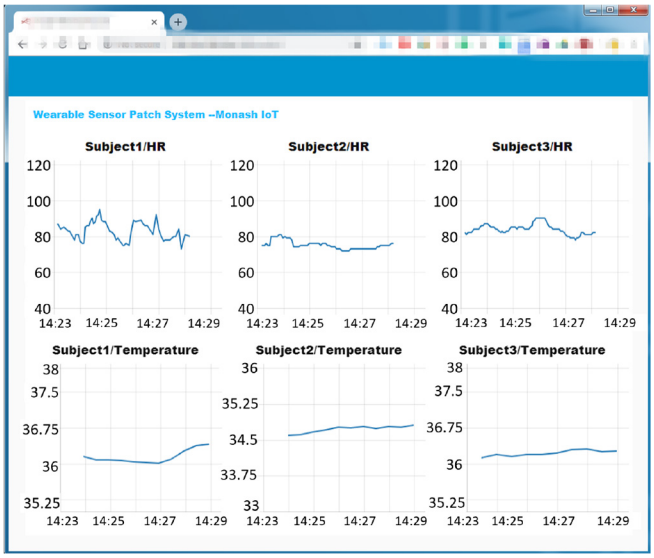


Fig. 14. Website UI on the cloud server for remote health monitoring of three subjects in the same room.

to the hand [20], [21] or chest [18], [22]. However, devices attached to hands can be easily influenced by daily movements. Gatouillat *et al.* [22] presented a chest-based sensor, while the belt around abdomen is not an optimal choice due to comfort consideration. In this article, a rigid-flex sensor patch design is proposed for health monitoring, which can be easily attached to people's chest without much disturbance to daily life. To extend the operation time of wearable devices, low-power circuits and cycled operation loop is necessary in the sensor design. The operation time of a wearable device is usually within tens of hours [64] to a few days [20]. In this article, a small rechargeable coin battery (120 mAh) is used as the power supply, which can last for 100 h in a 5-min operation loop.

For healthcare applications, the cardiorespiratory system is the one that most of the medical professionals pay attention to. Different designs of ECG and PPG sensors are common devices that can measure RR [19], [20], HR [18], [20]–[22], and HR variability [20]. In this article, both ECG and PPG sensors are integrated with the proposed sensor patch, which have been validated in our previous works [37], [43]. The ECG and PPG sensors are designed to be detached and assembled easily in different monitoring scenarios for consideration of power consumption and comfort. Besides that, the combination of ECG and PPG leads to another vital sigh, the estimation of BP by the PTT [14], [49]. The data collection, analysis, and evaluation of this BP estimation method are presented in our previous work [51].

For IoT-connected healthcare applications, wireless communication is essential for remote health monitoring scenarios. Due to the low-power consumption and wide usage of electronic devices, BLE is one of the most common wireless technologies in wearable healthcare devices. However, the transmission range of BLE is within 100 m, thus different gateways are proposed for long-range data transmission. Smartphones are selected in [21] and [22] for mobile gateways, while other portable computers, such as Raspberry Pi and SPHERE Gateway, are used in [20] and [64] for fixed gateways. In this article, both smartphones and portable computers are used to cover short- and long-range data transmission with BLE and Wi-Fi, respectively. The MQTT protocol is implemented in both gateways, which is a publish/subscribe messaging protocol suitable for low-data rate and low-power consumption IoT applications. To visualize the monitored health data, both a smartphone application and a Web application are designed for users and medical professionals respectively.

VI. Conclusion

In this article, an innovative low-power wearable sensor patch is proposed for IoT-connected remote long-term healthcare applications. The sensor patch system consists of three main parts: 1) a center board for signal acquisition, processing, and transmission; 2) a power board for energy supply and charging batteries; and 3) different sensors for physiological parameters measurements. All components of the sensor patch is connected and presented in a rigid-flex structure, which is suitable for wearable health monitoring of ECG, PPG, HR, and body temperature. As the ECG and PPG are integrated on the same device, continuous BP estimation based on the PAT method can be achieved without extra wires or hardware configurations. The experimental results demonstrate the performance of the proposed sensor patch against the comparison with a commercial reference medical equipment.

For IoT-connected healthcare applications, a BLE module is employed to transmit health data from the sensor patch to an IoT gateway, which will forward data to an Internet cloud server for storage and further analysis. Both fixed (Raspberry Pi) and mobile (smartphone) gateways are developed, thus the proposed sensor patch can operate continuously indoors and outdoors. Data encryption is employed in the sensor patch and the gateway for privacy protection, while a timestamp is paired with each group of health data the gateway receives for synchronization and integrity in the cloud. The real-time physiological measurements can be visualized on the cloud website for remote healthcare applications. In our future work, more edge computing functions on the gateway will be developed for an IoT-connected healthcare platform.

References

[1] I. Lee and K. Lee, "The Internet of Things (IoT): Applications, investments, and challenges for enterprises," *Bus. Horizons*, vol. 58, no. 4, pp. 431–440, 2015.

[2] F. Wu, T. Wu, and M. Yuce, "An Internet-of-Things (IoT) network system for connected safety and health monitoring applications," *Sensors*, vol. 19, no. 1, p. 21, 2019.

[3] C. C. Y. Poon, B. P. L. Lo, M. R. Yuce, A. Alomainy, and Y. Hao, "Body sensor networks: In the era of big data and beyond," *IEEE Rev. Biomed. Eng.*, vol. 8, no. 8, pp. 4–16, Apr. 2015. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/7096962

[4] U. E. Bauer, P. A. Briss, R. A. Goodman, and B. A. Bowman, "Prevention of chronic disease in the 21st century: Elimination of the leading preventable causes of premature death and disability in the USA," *Lancet*, vol. 384, no. 9937, pp. 45–52, 2014.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

[5] M. Elliott and A. Coventry, "Critical care: The eight vital signs of patient monitoring," *Brit. J. Nurs.*, vol. 21, no. 10, pp. 621–625, 2012.

[6] S. B. Baker, W. Xiang, and I. Atkinson, "Internet of Things for smart healthcare: Technologies, challenges, and opportunities," *IEEE Access*, vol. 5, pp. 26521–26544, 2017.

[7] S. Majumder, T. Mondal, and M. J. Deen, "Wearable sensors for remote health monitoring," *Sensors*, vol. 17, no. 1, p. 130, 2017.

[8] H. H. Asada, P. Shaltis, A. Reisner, S. Rhee, and R. C. Hutchinson, "Mobile monitoring with wearable photoplethysmographic biosensors," *IEEE Eng. Med. Biol. Mag.*, vol. 22, no. 3, pp. 28–40, May/Jun. 2003.

[9] Z. Zhu, T. Liu, G. Li, T. Li, and Y. Inoue, "Wearable sensor systems for infants," *Sensors*, vol. 15, no. 2, pp. 3721–3749, 2015.

[10] S. Milici, A. Lázaro, R. Villarino, D. Girbau, and M. Magnarosa, "Wireless wearable magnetometer-based sensor for sleep quality monitoring," *IEEE Sensors J.*, vol. 18, no. 5, pp. 2145–2152, Mar. 2018.

[11] H. Lin, W. Xu, N. Guan, D. Ji, Y. Wei, and W. Yi, "Noninvasive and continuous blood pressure monitoring using wearable body sensor networks," *IEEE Intell. Syst.*, vol. 30, no. 6, pp. 38–48, Nov./Dec. 2015.

[12] Y. Zheng, C. C. Poon, B. P. Yan, and J. Y. Lau, "Pulse arrival time based cuff-less and 24-H wearable blood pressure monitoring and its diagnostic value in hypertension," *J. Med. Syst.*, vol. 40, no. 9, p. 195, 2016.

[13] A. M. Carek, J. Conant, A. Joshi, H. Kang, and O. T. Inan, "SeismoWatch: Wearable cuffless blood pressure monitoring using pulse transit time," *Proc. ACM Interact. Mobile Wearable Ubiquitous Technol.*, vol. 1, no. 3, p. 40, 2017.

[14] S. S. Thomas, V. Nathan, C. Zong, K. Soundarapandian, X. Shi, and R. Jafari, "BioWatch: A noninvasive wrist-based blood pressure monitor that incorporates training techniques for posture and subject variability," *IEEE J. Biomed. Health Informat.*, vol. 20, no. 5, pp. 1291–1300, Sep. 2016.

[15] C. Buratti, A. Conti, D. Dardari, and R. Verdone, "An overview on wireless sensor networks technology and evolution," *Sensors*, vol. 9, no. 9, pp. 6869–6896, 2009.

[16] S. R. Islam, D. Kwak, M. H. Kabir, M. Hossain, and K.-S. Kwak, "The Internet of Things for health care: A comprehensive survey," *IEEE Access*, vol. 3, pp. 678–708, 2015.

[17] T. Wu, F. Wu, J.-M. Redouté, and M. R. Yuce, "An autonomous wireless body area network implementation towards IoT connected healthcare applications," *IEEE Access*, vol. 5, pp. 11413–11422, 2017.

[18] G. Yang *et al.*, "A health-IoT platform based on the integration of intelligent packaging, unobtrusive bio-sensor, and intelligent medicine box," *IEEE Trans. Ind. Informat.*, vol. 10, no. 4, pp. 2180–2191, Nov. 2014.

[19] T. N. Gia *et al.*, "Low-cost fog-assisted health-care IoT system with energy-efficient sensor nodes," in *Proc. 13th Int. Wireless Commun. Mobile Comput. Conf. (IWCMC)*, Valencia, Spain, 2017, pp. 1765–1770.

[20] C. Beach *et al.*, "An ultra low power personalizable wrist worn ECG monitor integrated with IoT infrastructure," *IEEE Access*, vol. 6, pp. 44010–44021, 2018.

[21] M. S. Mahmud, H. Wang, A. M. Esfar-E-Alam, and H. Fang, "A wireless health monitoring system using mobile phone accessories," *IEEE Internet Things J.*, vol. 4, no. 6, pp. 2009–2018, Dec. 2017.

[22] A. Gatouillat, B. Massot, Y. Badr, E. Sejdić, and C. Gehin, "Building IoT-enabled wearable medical devices: An application to a wearable, multiparametric, cardiorespiratory sensor," in *Proc. 11th Int. Joint Conf. Biomed. Eng. Syst. Technol. (BIOSTEC)*, vol. 1, 2018, pp. 109–118.

[23] M. A. Khan and K. Salah, "IoT security: Review, blockchain solutions, and open challenges," *Future Gener. Comput. Syst.*, vol. 82, pp. 395–411, May 2018.

[24] B. J. Mohd, T. Hayajneh, and A. V. Vasilakos, "A survey on lightweight block ciphers for low-resource devices: Comparative study and open issues," *J. Netw. Comput. Appl.*, vol. 58, pp. 73–93, Dec. 2015.

[25] Y. Liu, S. Tian, W. Hu, and C. Xing, "Design and statistical analysis of a new chaotic block cipher for wireless sensor networks," *Commun. Nonlinear Sci. Numer. Simulat.*, vol. 17, no. 8, pp. 3267–3278, 2012.

[26] D. Hong, J.-K. Lee, D.-C. Kim, D. Kwon, K. H. Ryu, and D.-G. Lee, "LEA: A 128-bit block cipher for fast encryption on common processors," in *Proc. Int. Workshop Inf. Security Appl.*, 2013, pp. 3–27.

[27] T. Suzaki, K. Minematsu, S. Morioka, and E. Kobayashi, "TWINE: A lightweight block cipher for multiple platforms," in *Proc. Int. Conf. Select. Areas Cryptography*, 2012, pp. 339–354.

[28] E. Biham, A. Biryukov, and A. Shamir, "Cryptanalysis of skipjack reduced to 31 rounds using impossible differentials," in *Proc. Int. Conf. Theory Appl. Cryptograph. Techn.*, 1999, pp. 12–23.

[29] C.-C. Lu and S.-Y. Tseng, "Integrated design of AES (advanced encryption standard) encrypter and decrypter," in *Proc. IEEE Int. Conf. Appl. Specific Syst. Archit. Process.*, 2002, pp. 277–285.

[30] J. S. Arteaga-Falconi, H. A. Osman, and A. E. Saddik, "ECG authentication for mobile devices," *IEEE Trans. Instrum. Meas.*, vol. 65, no. 3, pp. 591–600, Mar. 2016.

[31] T. Lemlouma, A. Rachedi, M. A. Chalouf, and S. A. Chellouche, "A new model for NGN pervasive e-Health services," in *Proc. 1st Int. Symp. Future Inf. Commun. Technol. Ubiquitous HealthCare (Ubi-HealthTech)*, 2013, pp. 1–5.

[32] A. Boulemtafes, A. Rachedi, and N. Badache, "A study of mobility support in wearable health monitoring systems: Design framework," in *Proc. 12th Int. Conf. Comput. Syst. Appl. (AICCSA)*, 2015, pp. 1–8.

[33] S. R. Moosavi *et al.*, "End-to-end security scheme for mobility enabled healthcare Internet of Things," *Future Gener. Comput. Syst.*, vol. 64, pp. 108–124, Nov. 2016.

[34] S. González-Valenzuela, M. Chen, and V. C. M. Leung, "Mobility support for health monitoring at home using wearable sensors," *IEEE Trans. Inf. Technol. Biomed.*, vol. 15, no. 4, pp. 539–549, Jul. 2011.

[35] H. Fotouhi, M. Alves, M. Z. Zamalloa, and A. Koubâa, "Reliable and fast hand-offs in low-power wireless networks," *IEEE Trans. Mobile Comput.*, vol. 13, no. 11, pp. 2620–2633, Nov. 2014.

[36] J. E. Luzuriaga, J. C. Cano, C. Calafate, P. Manzoni, M. Perez, and P. Boronat, "Handling mobility in IoT applications using the MQTT protocol," in *Proc. Internet Technol. Appl. (ITA)*, 2015, pp. 245–250.

[37] T. Wu, J.-M. Redouté, and M. Yuce, "A wearable, low-power, real-time ECG monitor for smart T-shirt and IoT healthcare applications," in *Proc. Adv. Body Area Netw. I*, 2019, pp. 165–173.

[38] Analog Devices. (Mar. 2017). *Single-Lead, Heart Rate Monitor Front End AD8232 Datasheet*. [Online]. Available: http://www.analog.com/media/en/technical-documentation/data-sheets/AD8232.pdf

[39] P. Xiaofei, W. Lei, Z. Hui, Q. Yajie, and H. Zhiliang, "A low-power portable ECG sensor interface with dry electrodes," *J. Semicond.*, vol. 34, no. 5, 2013, Art. no. 055002.

[40] J. Allen, "Photoplethysmography and its application in clinical physiological measurement," *Physiol. Meas.*, vol. 28, no. 3, p. R1, 2007.

[41] T. Tamura, Y. Maeda, M. Sekine, and M. Yoshida, "Wearable photoplethysmographic sensors—Past and present," *Electronics*, vol. 3, no. 2, pp. 282–302, 2014.

[42] J. Lee, K. Matsumura, K.-I. Yamakoshi, P. Rolfe, S. Tanaka, and T. Yamakoshi, "Comparison between red, green and blue light reflection photoplethysmography for heart rate monitoring during motion," in *Proc. 35th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2013, pp. 1724–1727.

[43] T. Wu, J.-M. Redouté, and M. R. Yuce, "A wearable wireless medical sensor network system towards Internet-of-patients," in *Proc. IEEE Sensors Conf.*, New Delhi, India, 2018, pp. 1–3.

[44] T. Jeon, B. Kim, M. Jeon, and B.-G. Lee, "Implementation of a portable device for real-time ECG signal analysis," *Biomed. Eng. Online*, vol. 13, no. 1, p. 160, Nov. 2014.

[45] E. Nemati, M. J. Deen, and T. Mondal, "A wireless wearable ECG sensor for long-term applications," *IEEE Commun. Mag.*, vol. 50, no. 1, pp. 36–43, Jan. 2012.

[46] K. Fox *et al.*, "Resting heart rate in cardiovascular disease," *J. Amer. College Cardiol.*, vol. 50, no. 9, pp. 823–830, 2007.

[47] X.-R. Ding *et al.*, "Continuous blood pressure measurement from invasive to unobtrusive: Celebration of 200th birth anniversary of Carl Ludwig," *IEEE J. Biomed. Health Informat.*, vol. 20, no. 6, pp. 1455–1465, Nov. 2016.

[48] R. Mukherjee, S. Ghosh, B. Gupta, and T. Chakravarty, "A literature review on current and proposed technologies of noninvasive blood pressure measurement," *Telemed. e-Health*, vol. 24, no. 3, pp. 185–193, 2018.

[49] D. Buxi, J.-M. Redouté, and M. R. Yuce, "A survey on signals and systems in ambulatory blood pressure monitoring using pulse transit time," *Physiol. Meas.*, vol. 36, no. 3, p. R1, 2015.

[50] J. Solà *et al.*, "Noninvasive and nonocclusive blood pressure estimation via a chest sensor," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 12, pp. 3505–3513, Mar. 2013.

[51] C. Qiu, T. Wu, J.-M. Redouté, and M. R. Yuce, "A wireless wearable sensor patch for the real-time estimation of continuous beat-to-beat blood pressure," in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2019, pp. 6842–6845.

[52] C. Gomez, J. Oller, and J. Paradells, "Overview and evaluation of Bluetooth low energy: An emerging low-power wireless technology," *Sensors*, vol. 12, no. 9, pp. 11734–11753, 2012.

[53] A. F. Harris, III, V. Khanna, G. Tuncay, R. Want, and R. Kravets, "Bluetooth low energy in dense IoT environments," *IEEE Commun. Mag.*, vol. 54, no. 12, pp. 30–36, 2016.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.

[54] D. Thangavel, X. Ma, A. Valera, H.-X. Tan, and C. K.-Y. Tan, "Performance evaluation of MQTT and CoAP via a common middleware," in *Proc. IEEE 9th Int. Conf. Intell. Sensors Sensor Netw. Inf. Process. (ISSNIP)*, 2014, pp. 1–6.
[55] A. Banks and R. Gupta, *MQTT Version 3.1.1*, OASIS Standard 29, 2014.
[56] R. A. Light, "MoSQuitto: Server and client implementation of the MQTT protocol," *J. Open Source Softw.*, vol. 2, no. 13, p. 265, 2017.
[57] P. P. Ray, "A survey on Internet of Things architectures," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 30, no. 3, pp. 291–319, 2018.
[58] M. Maksimović, V. Vujović, N. Davidović, V. Milošević, and B. Perišić, "Raspberry PI as Internet of Things hardware: Performances and constraints," *Design Issues*, vol. 3, no. 8, pp. 1–6, 2014.
[59] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet Things J.*, vol. 3, no. 5, pp. 637–646, Oct. 2016.
[60] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 4, pp. 2322–2358, 4th Quart., 2017.
[61] P. P. Ray, "A survey of IoT cloud platforms," *Future Comput. Informat. J.*, vol. 1, nos. 1–2, pp. 35–46, 2016.
[62] A. M. Rahmani *et al.*, "Exploiting smart e-Health gateways at the edge of healthcare Internet-of-Things: A fog computing approach," *Future Gener. Comput. Syst.*, vol. 78, pp. 641–658, Jan. 2018.
[63] W. J. Buchanan, S. Li, and R. Asif, "Lightweight cryptography methods," *J. Cyber Security Technol.*, vol. 1, nos. 3–4, pp. 187–201, 2017.
[64] V. K. Sarker, M. Jiang, T. N. Gia, A. Anzanpour, A. M. Rahmani, and P. Liljeberg, "Portable multipurpose bio-signal acquisition and wireless streaming device for wearables," in *Proc. IEEE Sensors Appl. Symp. (SAS)*, 2017, pp. 1–6.

Taiyang Wu (Student Member, IEEE) received the B.E. degree from Southeast University, Nanjing, China, in 2014, and the Ph.D. degree from the Department of Electrical and Computer Systems Engineering, Monash University, Melbourne, VIC, Australia, in 2019.

His main areas of research interest are wireless sensor network, wearable biomedical sensors, IoT-connected healthcare applications, and energy harvesting techniques.

Fan Wu (Student Member, IEEE) received the B.E. degree from Monash University, Melbourne, VIC, Australia, in 2015, where he is currently pursuing the Ph.D. degree with the Department of Electrical and Computer Systems Engineering.

He was a Research Assistant with the Engineering Department, Monash University from 2015 to 2017. His main areas of research interest are wireless sensor network, wearable sensors, energy harvesting, triboelectric nanogenerator, and IoT innovations.

Chunkai Qiu (Student Member, IEEE) received the B.E. degree from Monash University, Melbourne, VIC, Australia, in 2017, where he is currently pursuing the Ph.D. degree with the Department of Electrical and Computer Systems Engineering.

His main area of research interest is wearable devices and triboelectric nanogenerators.

Jean-Michel Redouté (Senior Member, IEEE) received the M.S. degree in electronics from the University of Antwerp, Antwerp, Belgium, in 1998, and the M.Eng. degree in electrical engineering from the University of Brussels, Brussels, Belgium, in 2001.

In August 2001, he started working with Alcatel Bell, Antwerp, where he was involved in the design of analog microelectronic circuits for telecommunications systems. In January 2005, he joined the University of Leuven, Leuven, Belgium, as a Ph.D. Research Assistant. In May 2009, he defended his Ph.D. titled "Design of EMI resisting analog integrated circuit." In September 2009, he started working with the Berkeley Wireless Research Center, University of California at Berkeley, Berkeley, CA, USA: this research was funded by the Belgian American Educational Foundation. In September 2010, he joined Monash University, Melbourne, VIC, Australia, as a Senior Lecturer. In July 2018, he started working with the University of Liège, Liège, Belgium, as an Associate Professor. His research is concentrated on miniaturized and low-power sensor interfaces, robust mixed-signal integrated circuit design with a high immunity to electromagnetic interference, biomedical (integrated and nonintegrated) circuit design, electromagnetic interference and compatibility, and integrated imagers.

Mehmet Rasit Yuce (Senior Member, IEEE) received the M.S. degree in electrical and computer engineering from the University of Florida, Gainesville, FL, USA, in 2001, and the Ph.D. degree in electrical and computer engineering from North Carolina State University, Raleigh, NC, USA, in December 2004.

He is an Associate Professor with the Department of Electrical and Computer Systems Engineering, Monash University, Melbourne, VIC, Australia. He was a Postdoctoral Researcher with the Electrical Engineering Department, University of California at Santa Cruz, Santa Cruz, CA, USA, in 2005. He was an Academic Member of the School of Electrical Engineering and Computer Science, University of Newcastle, Callaghan, NSW, Australia, until July 2011. His research interests include wearable devices, Internet of Things for healthcare, wireless implantable telemetry, wireless body area network, biosensors, integrated circuit technology dealing with digital, analog, and radio-frequency circuit designs for wireless, biomedical, and RF applications. He has published more than 180 technical articles in the above areas.

Dr. Yuce received the NASA Group Achievement Award in 2007 for developing an SOI transceiver, the Best Journal Paper Award in 2014 from the IEEE Microwave Theory and Techniques Society, and the Research Excellence Award in the Faculty of Engineering and Built Environment, University of Newcastle in 2010. He is a Topical Editor of the IEEE SENSORS JOURNAL, an Editor-in-Chief of *Sensors*, and a Guest Editor of the IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS in 2015.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2025 at 05:46:04 UTC from IEEE Xplore. Restrictions apply.